Quiz — 2.1.5 Thinking Concurrently

Name: ______________________ Date: __________ Mark: ______ / 20

OCR H446 · Component 02 · 2.1.5

Section A — Multiple choice (6 marks)

1. A weather model splits a country into regions and processes each region at the same time on a multi-core machine. This is an example of: A. Caching B. Concurrent (parallel) processing C. A precondition D. Abstraction Your answer: ☐

2. Which pair of tasks in a video-editing app are **safe to run concurrently** because they are independent? A. Rendering a clip and then exporting the very same rendered clip B. Generating audio for scene 1 and generating audio for scene 2 C. Reading a value, then writing back that value plus one D. Logging in, then loading the logged-in user's projects Your answer: ☐

3. Two concurrent threads both update a shared bank balance without coordination, and the final figure depends on which finishes first. This is a: A. Precondition B. Race condition C. Reusable component D. Decision point Your answer: ☐

4. On a computer with only a **single core**, running tasks "concurrently" actually means: A. The tasks run literally at the same instant B. The processor time-slices, rapidly switching between tasks C. The tasks are cached D. Only one task can ever be started Your answer: ☐

5. A key benefit of doing a long file download concurrently with the rest of a program is that: A. The download is guaranteed to finish instantly B. The user interface stays responsive while the download runs C. Race conditions become impossible D. The program no longer needs any inputs Your answer: ☐

6. Two processes each hold a resource the other needs and neither will release it, so both wait forever. This is: A. Caching B. Deadlock C. Decomposition D. A precondition Your answer: ☐

Section B — Short answer (9 marks)

7. An online store generates a sales report by (i) totalling each region's sales and (ii) combining the regional totals into a grand total. Explain which part could run concurrently and which must run afterwards, referring to data dependency. **[3]**

8. Discuss **one** benefit and **one** trade-off of using concurrency to load the images on a busy news website. **[2]**

9. Explain why running a task concurrently is **not always faster**, giving one reason. **[2]**

10. State what a *race condition* is and give one example of where it could occur in a shared online ticket-booking system. **[2]**

Section C — Applied question (5 marks)

11. A photo-sharing app must, when a user uploads a photo: (a) generate three different thumbnail sizes, (b) scan the image for prohibited content, and (c) save the original to storage. Using *thinking concurrently*, identify which of these tasks could run at the same time and which must be sequenced, then discuss **one** benefit and **one** trade-off of running them concurrently. **[5]**